FILE HANDLING DEMO

1. Aim

To write a sentence into a text file, read the file contents, convert the text to lowercase, and count the occurrences of the character pairs **“pe”** and **“pi”**.

2. Description

This Java program demonstrates basic file handling. It uses **FileWriter** to create the file **sample.txt** and write a sentence into it. The program then uses **File** and **Scanner** to read the text from the file. After converting the text to lowercase, it checks each pair of adjacent characters and counts the pairs “pe” and “pi”. Finally, it displays both counts. Any file-related error is handled using an **IOException** catch block.

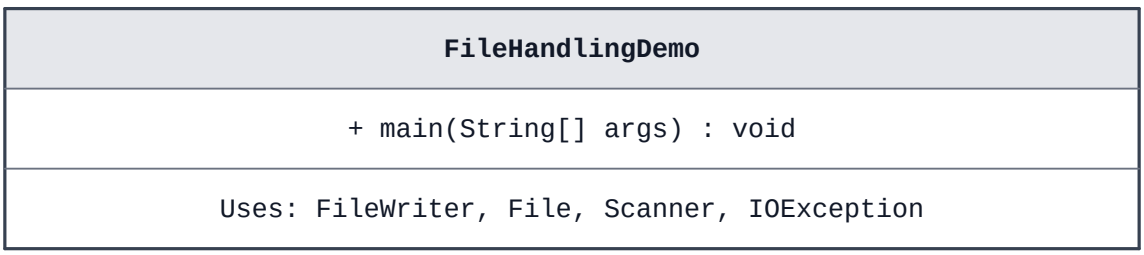
3. Software Requirements

• Programming Language: Java
• JDK: Java Development Kit (JDK 8 or later)
• IDE/Text Editor: Eclipse, IntelliJ IDEA, VS Code, or any Java editor
• Operating System: Windows, Linux, or macOS

4. Algorithm

1. Start the program.
2. Store the given sentence in a String variable.
3. Create sample.txt using FileWriter and write the sentence into it.
4. Close the FileWriter and display a success message.
5. Open sample.txt using File and Scanner.
6. Read all lines and combine them into one String.
7. Close the Scanner and convert the content to lowercase.
8. Initialize the counters pe and pi to zero.
9. Examine each pair of adjacent characters in the content.
10. If the pair is “pe”, increment pe; otherwise, if it is “pi”, increment pi.
11. Display the two counts.
12. If an IOException occurs, display an error message.
13. Stop the program.

5. UML Diagram



6. Code

```
package recordprograms;

import java.io.File;
import java.io.FileWriter;
import java.io.IOException;
import java.util.Scanner;

public class FileHandlingDemo
{
    public static void main(String[] args)
    {
        String sentence =
            "Peter Piper picked a peck of pickled peppers. "
            + "A peck of pickled peppers Peter Piper picked. "
            + "If Peter Piper picked a peck of pickled peppers, "
            + "where is the peck of pickled peppers Peter Piper picked?";

        try
        {
            FileWriter fw = new FileWriter("sample.txt");
            fw.write(sentence);
            fw.close();

            System.out.println("File created and data written successfully.");

            File f = new File("sample.txt");
            Scanner sc = new Scanner(f);

            String content = "";

            while (sc.hasNextLine())
            {
                content = content + sc.nextLine() + " ";
            }

            sc.close();

            content = content.toLowerCase();

            int pe = 0;
            int pi = 0;

            for (int i = 0; i < content.length() - 1; i++)
            {
                String pair = content.substring(i, i + 2);

                if (pair.equals("pe"))
                    pe++;

                else if (pair.equals("pi"))
                    pi++;
            }

            System.out.println("Count of pe: " + pe);
            System.out.println("Count of pi: " + pi);
        }
        catch (IOException e)
        {
            System.out.println("Error while handling file.");
        }
    }
}
```

7. Output Screenshot

```
File created and data written successfully.
Count of pe: 12
Count of pi: 6
```

8. Result

Thus, the Java file-handling program was executed successfully. It created the text file, wrote the sentence, read the file contents, and displayed the counts of “pe” and “pi”.